

theory. Besides, a manager would question whether such non-randomness can lead to any meaningful profitable rules. If I need to gamble \$1 on 10,000 runs and expect to make \$1 for my efforts, then I would do much better in the part-time employment of a janitorial agency.

But the result bears another suspicious element. Of more practical relevance here is the following severe problem about non-randomness. Even the fathers of statistical science forgot that a random series of runs need not exhibit a pattern to look random; as a matter of fact data that is perfectly patternless would be extremely suspicious and appear to be man-made. A single random run is bound to exhibit some pattern – if one looks hard enough. Note that Professor Pearson was among the first scholars who were interested in creating artificial random data generators, tables one could use as inputs for various scientific and engineering simulations (the precursors of our Monte Carlo simulator). The problem is that they did not want these tables to exhibit any form of regularity. Yet real randomness does not look random!

I would further illustrate the point with the study of a phenomenon well known as cancer clusters. Consider a square with 16 random darts hitting it with equal probability of being at any place in the square. If we divide the square into 16 smaller squares, it is expected that each smaller square will contain one dart on average – but only on average. There is a very small probability of having exactly 16 darts in 16 different squares. The average grid will have more than one dart in a few squares, and no dart at all in many squares. It will be an exceptionally rare incident that no (cancer) cluster would show on the grid. Now, transpose our grid with the darts in it to overlay a map of any region. Some newspaper will declare that one of the areas (the one with more than the average of darts) harbors radiation that causes cancer, prompting lawyers to start soliciting the patients.

THE DOG THAT DID NOT BARK: ON BIASES IN SCIENTIFIC KNOWLEDGE

By the same argument, science is marred by a pernicious survivorship bias, affecting the way research gets published. In a way that is similar to journalism, research that yields no result does not make it to print.

That may seem sensible, as newspapers do not have to have a screaming headline saying that nothing new is taking place (though the Bible was smart enough to declare *ein chadash tacht hashemesh* – “nothing new under the sun”, providing the information that things just do recur). The problem is that a finding of absence and an absence of findings get mixed together. There may be great information in the fact that *nothing took place*. As Sherlock Holmes noted in the *Silver Blaze* case – the curious thing was that the dog did not bark. More problematically, there are plenty of scientific results that are left out of publications because they are not statistically significant, but nevertheless provide information.⁹

I Have No Conclusion

I am frequently asked the question: when is it truly not luck? To be honest, I am unable to answer it. I can tell that person A seems less lucky than person B, but the confidence in such knowledge can be so weak as to be meaningless. I prefer to remain a skeptic. People frequently misinterpret my opinion. I never said that every rich man is an idiot and every unsuccessful person unlucky, only that in absence of much additional information I prefer to reserve my judgment. It is safer.